

Where Does Physics Help a Graph Network?

A Leakage-Controlled Study of SEIR-Informed Spatio-Temporal GNNs for Dengue Forecasting

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ABSTRACT

Spatio-temporal graph neural networks (ST-GNNs) and SEIR-informed neural models are both proposed for district-level dengue forecasting, but they are rarely compared under one leak-free protocol. We study whether replacing the LSTM of an SEIR-informed forecaster with a graph encoder helps, on Sri Lanka’s weekly record (25 districts, three-week horizon). Before modelling, we audit the benchmark array used by prior work: its rows are out of date order, every training split contains 2023 data from after the test periods, its case and weather columns are on different timelines, its “lag 12” rainfall channel holds rain from twelve weeks *ahead*, and its largest spike is a spreadsheet error. We rebuild a 559-week dataset from hash-verified original sources, and one self-contained notebook reproduces every result from them. On one harness, six encoders (five published ST-GNNs and the SEIR-LSTM’s LSTM) are each paired with a direct head, a pure SEIR decoder and a gated SEIR head. Architecture matters little: the working encoders span 0.20 RMSE and the graph adds 0.06. The pure SEIR decoder is worse than persistence on every encoder, and 7.3–7.4 RMSE worse than the direct head on each working one, for structural reasons: 14–16% of targets lie below its zero-transmission floor and its force of infection is only $r^2 \approx 0.25$ predictable. A gated SEIR head, which lets the simulator supply only the departure from persistence, brings all six encoders below persistence and repairs the three that fail as published by 5.3–17.0 validation RMSE on 9/9 paired runs. A pre-registered control shows that this repair is not physics: the same head without the simulator accounts for 96–112% of it, and adding the simulator is significantly worse on one failing encoder, not better on the other two, and 0.75–0.86 worse on the working ones. A negative-binomial likelihood is the largest controlled gain (−0.76, 9/9). The graph beats the LSTM inside the gated SEIR head on three origins, and that advantage is not robust on nine disjoint origins. More than twenty further levers are not adopted, and the working models make the same errors (residual correlation 0.91–0.97). On this target the useful prior is persistence, not transmission dynamics.

KEYWORDS

dengue forecasting, spatio-temporal graph neural networks, SEIR, physics-informed learning, data leakage, reproducibility, negative results

1 INTRODUCTION

Dengue is the fastest-spreading mosquito-borne viral disease, and Sri Lanka faces recurring severe epidemics: the 2017 outbreak alone recorded over 186,000 cases and 440 deaths [15]. Roughly three weeks separate infection from diagnosis. District-level forecasts issued $H = 3$ weeks ahead therefore carry actionable lead time for clinical capacity and vector control.

Two families of model are proposed for this task. Spatio-temporal graph neural networks, transferred from traffic forecasting, treat districts as nodes and learn from neighbours; Weng et al. [17] benchmark five of them on Sri Lanka’s record. SEIR-informed models push the forecast through compartmental dynamics; Liu et al. [7] drive an SEIR model with an LSTM on the same country. The natural hybrid is an *SEIR-GNN*: the SEIR-LSTM with its LSTM replaced by a graph encoder, so that spatial coupling and mechanism act together. Whether that hybrid helps, and through which component, is the question of this paper.

One property of the data governs every answer: weekly incidence is strongly autocorrelated. On the corrected data, cases at $t - 1$ explain $r^2 = 0.89$ of cases at t , while the best single climate covariate at a causal lag explains 0.03. Every model competes with the naive persistence forecast $\hat{y}_{t+h} = y_t$. A benchmark that omits it cannot tell whether a model contributed anything, and the reference benchmark omits it.

Our first SEIR-GNN pipeline reported a win, and it was wrong. Its headline set the test error at one forecast origin (26.1) against persistence averaged over three (36.0); at that same origin persistence scored 22.4, and it won at every origin. The pipeline also trained for about sixty gradient steps, optimised SMAPE while scoring RMSE, held the force of infection constant across the horizon, and collapsed every covariate to a scalar before the graph saw it. This paper reports the controlled re-run. Its protocol and each experiment’s decision rule were committed before the runs, and negative results are reported as such.

Contributions.

- **A data audit and a corrected dataset** (§3). The benchmark array mixes time orders, puts post-test data in every training split, and feeds future climate as “lagged” features. We rebuild 559 weeks from source reports, with causal lags enforced by perturbation tests.
- **One harness for six encoders and four SEIR-informed heads** (§4), with validation-only selection, paired tests at two units, and a nine-origin confirmation.
- **Whether physics helps** (§5). A pure SEIR decoder fails on every encoder, for reasons we measure. A gated SEIR head appears to repair the three encoders that fail as published, but a pre-registered control attributes 96–112% of that repair to the persistence anchor and none of it to the simulator (§5.2).
- **An informational ceiling, measured** (§5.6). More than twenty architectural, data and physics levers leave the same errors behind (residual correlation 0.91–0.97 among the working models).
- **One notebook, from original sources.** Every number in this paper is produced by a single notebook that verifies the source files, rebuilds the data, defines every model in plain PyTorch, trains all 810 runs and analyses only those runs.

2 RELATED WORK

ST-GNNs for dengue. Spatio-temporal graph networks couple a graph operator over regions with a temporal operator over history. The five we evaluate come from traffic forecasting. STGAT and A3TGCN [1] use attention, ASTGCN [4] attends over space and time, AAGCN [14] adapts its adjacency, and DCRNN [6] diffuses over the graph inside a recurrent cell. Weng et al. [17] benchmark all five on Sri Lanka’s dengue record, and GulMohamed et al. [3] add per-horizon heads, temporal attention and a Gaussian output.

Mechanistic hybrids. Liu et al. [7] forecast Sri Lankan dengue with an SEIR model whose force of infection is predicted by an LSTM; we call this SEIR-LSTM. For epidemics more generally, EINNs [11] couple a neural forecaster with a mechanistic model through shared latent dynamics. CausalGNN [16] and MepoGNN [2] embed a metapopulation SEIR in a graph network, so that districts are coupled inside the dynamics. Physics-informed networks [10] add dynamics as a loss; MP-PINN [8] applies this to epidemic curves in a regime where data are scarce relative to the dynamics. We use the SEIR–SEI stage durations of Phaijoo and Gurung [9].

Baselines and remedies from forecasting. Simple baselines often match deep spatio-temporal models: STID [13] replaces graph operators with identity embeddings. Reversible instance normalisation [5] targets distribution shift, and count likelihoods such as the negative binomial of DeepAR [12] target skewed demand. TimeGAN [18] generates synthetic sequences for data-poor settings. We test each of these here, because each addresses a weakness our diagnosis found.

Table 1: Audit of the benchmark array, recomputed by the reproduction notebook from the original files.

finding	consequence
Rows 0–47 are 2023; rows 51 onward run 2013–2022	45 rows from 2023 sit in the training split of every fold, which tests on 2018–2022
Cases reordered, weather not	the array’s temperature matches ERA5 at $r = 0.95$ on its own row order and $r = 0.45$ on the dates the case rows carry
“Lag 12” precipitation	matches ERA5 rain 12 weeks <i>later</i> ($r = 0.68$) and not 12 weeks <i>earlier</i> ($r = 0.19$): future information
48 reports dropped, one week stored twice	454 of 459 rows equal a published report exactly
Week-395 “19× spike”	row A of one printed table is a dragged formula (15 of 24 cells are the sum of the two before, districts sum to 7,165 against a national cell of 35); the same table’s cumulative row gives the true count

3 DATA

3.1 Reproducing the benchmark

We ran the reference implementation [17] unmodified—their code, their array, their protocol—on a pinned software stack. It recovers their published numbers to within 7.3% across twenty comparisons, with AAGCN and DCRNN matching to four decimal places. The published results are real, but three properties of the evaluation account for the reported advantage. The column labelled “cross validated” is computed on a loader spanning the full series, so it includes training data. RMSE is averaged per window rather than pooled, which on this target is ~39% lower. No naive baseline is reported, and adding persistence to that same column reverses the ranking: it beats *every* model on MAE and RMSE. An independent re-implementation that shares no code with either codebase reproduces this. This reproduction runs the authors’ own code in its pinned environment and is packaged separately (reproduction/ in the repository); everything from §3.2 on is produced by the reproduction notebook.

3.2 Auditing the benchmark array

The benchmark array (`sri_lanka_2013–2022_shifted.npy`) has no date axis. We dated every row by matching its 25 district counts exactly against the weekly reports, and dated the weather channels separately by finding the calendar offset at which the array’s temperature best matches ERA5. Table 1 lists what that shows.

Two of these are information leaks. Training on 2023 while testing on 2018–2022 lets a model see the future of its test period, and a “lagged” rainfall channel that holds rain from twelve weeks ahead gives it that future’s weather. Any score on this array, including the published ones and our own earlier work, is therefore an optimistic bound. Earlier drafts of this study called the week-395 spike an administrative reporting backlog. It is a spreadsheet error, and it can be corrected from the same published table.

3.3 The corrected dataset

Every input is an original file, checked against the SHA-256 recorded when it was first retrieved: the parsed Table 1 rows of every Weekly Epidemiological Report, the one report PDF used for the correction, ERA5 daily responses from the Open-Meteo archive, and the Department of Census and Statistics population and census PDFs. From these the case series is rebuilt, keyed by report volume and number: 559 weeks from 2013-W26 to 2024-W10. The seven weeks with no report stay missing and are never interpolated; districts plus the Kalmunai division equal the Epidemiology Unit’s own national total in 432 of 552 reports. Weekly ERA5 temperature, precipitation, humidity and soil moisture are aggregated from daily values at an interior point of each district, and each week’s population is the latest official figure for an earlier year (the 2012 Census, whose 25 district totals sum exactly to the national total, then mid-year estimates).

Lags are applied in the model, not baked into the data: a forecast for week t reads cases up to $t - 1$ and climate up to $t - 2$, since ERA5 is published about five days late. On this series the persistence test RMSE on the frozen three origins (averaged over origins, as for every model) is **36.02**, against 44.80 on the array (29.52 with the week-395 windows excluded). The last test fold covers the 2023 epidemic.

4 MODELS AND PROTOCOL

A run is a tuple (encoder, head, likelihood, features). Each element is varied while the others are held fixed, so every difference is attributable to one component.

4.1 Encoders

The five published ST-GNNs (STGAT, A3TGCN, ASTGCN, AAGCN, DCRNN) are built from the reference implementations. The sixth encoder is the two-layer LSTM of the SEIR-LSTM [7]. Each maps a $W = 3$ week window of log cases, plus any exogenous features, to a 64-dimensional representation h_i per district. The graph is the benchmark’s district adjacency with self-loops, row-normalised; AAGCN runs with its adaptive adjacency off, as published.

4.2 Heads

All heads output fold-standardised $\log(1 + y)$ for $h = 1, 2, 3$. Let p_i be the persistence anchor (last observed week) in the same space and $f(h_i)$ a linear read-out.

- **direct**: $\hat{z}_i = f(h_i)$, as in the published ST-GNNs.
- **residual**: $\hat{z}_i = p_i + f(h_i)$.
- **gated**: $\hat{z}_i = p_i + \sigma(a)(f(h_i) - p_i)$, a single learned gate initialised at $\sigma(-2) = 0.12$. This head was added for §5.2; it is the gated SEIR head with the simulator removed.
- **SEIR decoder** (foi): $f(h_i)$ gives a weekly force of infection $\lambda_{i,h} = \exp(f(h_i) + b)$, clamped at $1/7$ per day. An SEIR model is stepped daily from an initial state seeded by observed cases: E_0 from cases at $t - 2$ scaled by the reporting rate $\rho = 1/11$, $S_0 = 1 - 0.682$ from a 2013–14 Colombo sero-survey, incubation rate $0.1/\text{day}$ and recovery rate $1/7/\text{day}$. Predicted incidence times ρ times population gives counts. A learned scale on E_0 and ρ is allowed.

- **gated SEIR** (foi_res): $\hat{z}_i = p_i + \sigma(a)(\text{SEIR}(\lambda_i) - p_i)$. The simulator supplies the *departure* from persistence. This is our SEIR-GNN when the encoder is a graph network, and SEIR-LSTM when it is the LSTM.
- **metapopulation SEIR** (foi_meta): the encoder predicts a transmission rate $\beta_i(t)$, and the simulator recomputes the force of infection each day as $\beta_i \sum_j C_{ij} I_j / N_j$, with $C = \text{softmax}(\log A + \Theta)$ a learned row-stochastic coupling initialised at the border graph A , as in MepoGNN and CausalGNN [2, 16]. It is gated like foi_res.

4.3 Likelihood, features and physics penalty

The default objective is squared error on the standardised target. The negative-binomial (NB) alternative adds a dispersion head and minimises the NB2 negative log-likelihood with the mean $\mu = \exp(\hat{z}\sigma + m) - 1$. Under squared error on $\log(1 + y)$, exp of the fitted value is a conditional *median*, while RMSE rewards the conditional *mean*; NB’s μ is the mean. Seasonal features are sin and cos of the week of year at one and two cycles per year. The spatial physics penalty adds $\lambda_s \sum_{(i,j) \in E} (\hat{y}_i/s_i - \hat{y}_j/s_j)^2$ with s_i the district’s training-mean cases: relative incidence should not jump across a border.

4.4 Protocol

Rolling origins at 0.55, 0.70 and 0.85 of the series, three seeds each. The 30 windows before each origin are validation, the following 15% are test, and normalisation uses training weeks only. Windows touching a missing week are dropped, never imputed. Training uses Adam (learning rate 3×10^{-3} , weight decay 10^{-4} , batch 32), early stopping on validation with patience 40, and 300–400 epochs. Each run’s RMSE is pooled over its windows on raw counts, overall and per horizon; tables report the mean over runs.

Selection uses validation RMSE only. Test is written to disk and never used to choose. The protocol, and for each later experiment its arms, control, decision rule and a prediction of the outcome, were committed before it ran. An arm is adopted if its mean validation delta against the control is negative, it wins at least 7 of 9 (origin, seed) units, and it does not raise horizon-3 error.

Two pairing units. Differences are paired on matched (origin, seed) units and tested with an exact two-sided sign-flip permutation test, BH-adjusted within each family. The *origin* unit (seeds averaged) is what a claim about the series rests on, but with three origins its p -value cannot fall below 0.25. The (origin, seed) unit reaches 0.004 but mostly measures run-to-run stability. Claims about the series therefore need the **nine-origin** protocol: origins at $0.40 + k/15$ ($k = 0, \dots, 8$), with nine disjoint test spans of $1/15$ of the series, where the origin unit reaches 0.004.

Reproducibility. One notebook produces every number in this paper. It reads only original source files, each verified against the SHA-256 recorded at retrieval; rebuilds the data; defines the five published architectures and the LSTM in plain PyTorch (each checked weight-for-weight against the reference implementation); and trains every configuration once, in one environment: 69 configurations on the three origins and 7 on the nine, three seeds each, 810 runs in 3.0 hours on four Kaggle CPU cores. A configuration shared by several comparisons, such as B , is trained once, so every

Table 2: The six encoders on one harness (corrected data, 3 origins $\times$ 3 seeds, squared error, no seasonal features). Mean validation / test RMSE. The pure SEIR decoder (foi) is worse than persistence on every encoder; the gated SEIR head (foi_res) brings all six close together, below persistence.

encoder	direct	SEIR decoder	gated SEIR
AAGCN	16.71 / 35.15	24.01 / 49.54	17.43 / 35.58
ASTGCN	16.83 / 35.61	24.21 / 51.75	17.51 / 35.71
LSTM	16.91 / 35.10	24.32 / 50.26	17.69 / 35.52
STGAT	22.93 / 62.00	25.53 / 52.37	17.63 / 35.84
A3TGCN	28.63 / 60.40	25.55 / 52.92	17.68 / 35.50
DCRNN	34.64 / 73.52	25.75 / 52.26	17.67 / 35.52
persistence		17.86 / 36.02	

pair is matched on the same runs. Runs are single-threaded and reproducible from their seeds.

5 RESULTS

All numbers are RMSE in weekly cases on the corrected dataset, from the 810 runs of the reproduction notebook. Validation decides and test is reported beside it. Unless stated otherwise, pairs are on the (origin, seed) unit of the frozen three origins.

5.1 Six encoders, one harness

Table 2 and Figure 1 give the comparison the SEIR-GNN question needs, with every encoder sharing folds, loss, early stopping and metric.

Architecture matters little, and the graph less. Behind the direct head AAGCN, ASTGCN and the LSTM score 16.71, 16.83 and 16.91: a spread of 0.20, and all three beat persistence (17.86). The LSTM has no message passing at all. A GCN beats the same network without message passing by only 0.06 (3/9, Table 3). STGAT, A3TGCN and DCRNN score 22.93, 28.63 and 34.64, all worse than persistence. A linear model fitted to their training residuals recovers 48% and 60% of STGAT’s and A3TGCN’s validation residual variance, but only 3–7% of the working encoders’. The failing encoders are underfitting, limited by their wiring rather than by the data.

The SEIR decoder fails on every encoder. Routing the forecast entirely through the simulator scores 24.01–25.75, worse than persistence on all six and 7.3–7.4 worse than the direct head on each working encoder. The deficit is structural, and we measured it without training by inverting the simulator for the force of infection that reproduces each observed count: (i) with transmission switched off, the E already seeded from recent cases still emits *more* cases than observed for 14–16% of training targets, which are therefore unreachable at any network output (39–41% of cells need a force of infection of zero); (ii) the required force of infection is only $r^2 = 0.22$ –0.26 predictable from case history, against 0.81–0.82 for the direct target; (iii) it is not susceptible depletion, since only 0.7–4.9% of cells reach the susceptible clamp.

The gated SEIR head repairs the failing encoders. Letting the simulator supply only the departure from persistence brings all six encoders to 17.43–17.69, below persistence on validation (17.86) and on test (35.50–35.84 against 36.02). For the three failing

Table 3: Every lever, paired against its own control on matched (origin, seed) runs. Δ is validation RMSE (negative is better); wins count the 9 units where the arm is better; p_{adj} is BH across this table. Test Δ is shown and decided nothing. With three origins these p -values describe run-to-run stability, not the series (§4.4).

lever	Δ val	wins	p_{adj}	Δ test
<i>Worked</i>				
NB likelihood (vs squared error)	-0.76	9/9	0.014	+1.61
Seasonal features	-0.76	6/9	0.185	+0.70
Best model B vs persistence	-2.35	9/9	0.014	+1.53
<i>Graph and architecture</i>				
Graph (GCN vs no message passing)	-0.06	3/9	0.501	-0.07
Learned adjacency	-0.14	5/9	0.185	-0.33
District seasonal curves	-0.05	6/9	0.757	+0.11
Nonlinear (MLP) head	+0.28	1/9	0.027	-0.31
MLP head + climate lags 2–13	+0.13	3/9	0.482	-0.70
Global context node	+0.02	3/9	0.844	-1.40
Residual head + NB	+0.47	2/9	0.207	-2.96
Combined architecture changes	+0.55	2/9	0.078	+0.59
<i>Literature remedies</i>				
RevIN	+1.47	0/9	0.014	-0.42
District identity embedding	+0.19	3/9	0.219	+0.98
STID-style MLP	+0.84	0/9	0.014	-1.22
NB-GLM (no network)	+1.46	0/9	0.014	+5.31
k -NN analogues (no network)	+1.01	0/9	0.014	-3.20
Ensemble B + NB-GLM + k -NN	+0.06	4/9	0.501	-1.79
<i>Data</i>				
Half the training windows	+0.39	2/9	0.050	+0.47
TimeGAN augmentation	+0.60	0/9	0.014	+0.52
SEIR-simulated pre-training	+0.03	6/9	0.822	+0.92
Climate, lags 2–4	+0.18	2/9	0.199	+0.51
Climate, lags 2–13	+0.19	2/9	0.302	-0.05
Climate, lags 2–25	+0.47	0/9	0.014	-0.47
<i>Physics</i>				
SEIR as auxiliary loss	+0.02	4/9	0.812	-0.32
SEIR simulator vs no-physics twin (AAGCN)	+0.86	1/9	0.020	+2.05
Spatial physics penalty	-0.06	6/9	0.207	-0.02
Metapopulation vs gated SEIR head	+0.20	1/9	0.020	+0.04
Gated SEIR-GNN vs SEIR-LSTM	-0.14	8/9	0.020	+0.02

encoders this is a large repair, on 9/9 paired runs each: STGAT –5.30 (test –26.16), A3TGCN –10.95 (–24.90) and DCRNN –16.97 (–38.01). It costs the working encoders 0.68–0.78 on validation. Read at face value, this is the physics-informed result the study set out to find. The next section shows that it is not.

5.2 Anchor, gate or physics?

The gated SEIR head differs from the direct head in three ways at once: it forecasts a departure from last week’s value (the anchor), it shrinks that departure through a learned gate, and the departure comes from the SEIR simulator. Only the third is physics. A pre-registered control separates them by training all six encoders behind four heads that add these one at a time: direct, residual (anchor), gated (anchor and gate) and gated SEIR (all three). The rescue is credited to physics only if gated SEIR beats its no-physics twin on at least 7 of 9 runs with $p_{\text{adj}} < 0.05$ on each failing encoder. Our

Table 4: The best model and the SEIR-GNN against their controls. Three-origin rows pair 9 (origin, seed) runs; nine-origin rows pair the 9 disjoint origins (seeds averaged), the unit a claim about the series rests on (attainable p floor 0.004). Raw two-sided p .

comparison	validation			test		
	Δ	wins	p	Δ	wins	p
B vs persistence, 3 origins	-2.35	9/9	0.004	+1.53	3/9	0.230
B vs persistence, 9 origins	-2.58	8/9	0.105	+2.24	4/9	0.293
gated SEIR-GNN (ASTGCN) vs SEIR-LSTM, 3 origins	-0.54	9/9	0.004	+0.14	5/9	0.594
gated SEIR-GNN (AAGCN) vs SEIR-LSTM, 3 origins	-0.14	8/9	0.008	+0.02	5/9	0.855
AAGCN vs LSTM, direct head, 3 origins	-0.27	6/9	0.082	+1.96	1/9	0.012
gated SEIR-GNN (ASTGCN) vs SEIR-LSTM, 9 origins	-0.62	8/9	0.012	+0.07	6/9	0.883
gated SEIR-GNN (AAGCN) vs SEIR-LSTM, 9 origins	+0.18	6/9	0.797	-0.30	5/9	0.078
metapopulation SEIR-GNN vs SEIR-LSTM, 9 origins	+0.49	4/9	0.398	-0.01	4/9	0.961
AAGCN vs LSTM, direct head, 9 origins	-0.72	6/9	0.516	-0.88	5/9	0.820
gated SEIR-GNN (AAGCN) vs persistence, 9 origins	-1.98	8/9	0.137	-0.59	6/9	0.680

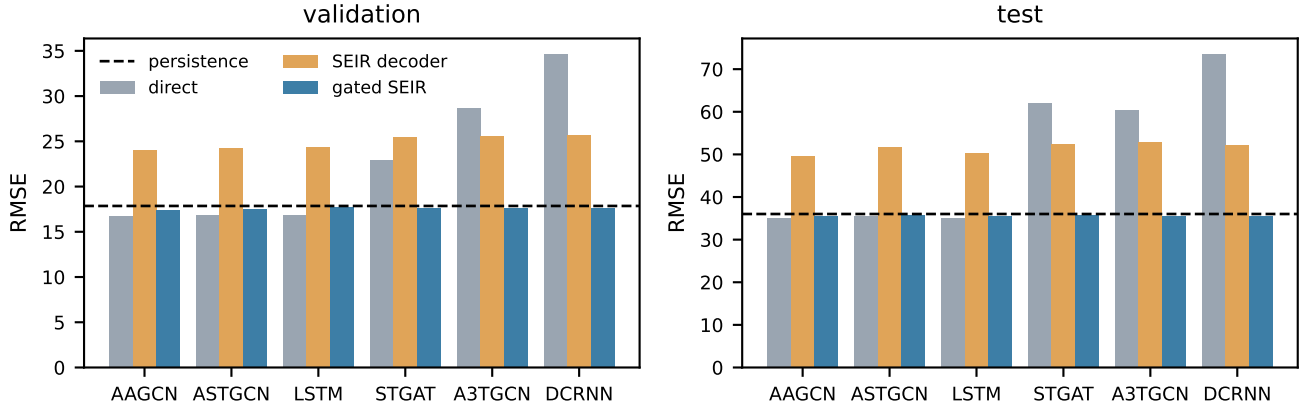


Figure 1: Six encoders behind three heads (Table 2). The direct head separates encoders that work from encoders that fail; the SEIR decoder is worse than persistence on all of them; the gated SEIR head puts all six at the same level, below persistence (dashed) on validation and test.

Table 5: Anchor, gate or physics? Each encoder behind four heads that add, in turn, the persistence anchor, the learned gate and the SEIR simulator (mean validation RMSE; configuration of Table 2). The last two columns pair the gated SEIR head with its no-physics twin (gated); negative favours the physics, and the count is runs where it wins.

encoder	direct		gated		SEIR – gated	
			resid.	gated	val	test
AAGCN	16.71	16.72	16.57	17.43	+0.86 (1/9)	+2.05 (0/9)
ASTGCN	16.83	16.85	16.76	17.51	+0.75 (0/9)	+0.95 (3/9)
LSTM	16.91	16.96	16.94	17.69	+0.75 (0/9)	-0.33 (6/9)
STGAT	22.93	17.85	16.98	17.63	+0.66 (1/9)	-0.62 (8/9)
A3TGCN	28.63	17.47	17.68	17.68	-0.00 (6/9)	-1.66 (9/9)
DCRNN	34.64	17.52	18.35	17.67	-0.68 (3/9)	-3.50 (9/9)
persistence			17.86			

stated prediction was that the anchor and gate carry most of the rescue.

Table 5 gives the result: **the rescue is the anchor, not the physics**. Anchor and gate alone account for 96–112% of the validation repair on each failing encoder, and a plain residual head, with no gate and no simulator, already brings them to 17.47–17.85. Adding the simulator to the gated head earns credit on none of the three: it is significantly worse on STGAT (+0.66, 1/9, $p_{\text{adj}} = 0.023$), ties on A3TGCN (−0.00, 6/9), and helps DCRNN by −0.68 on only 3/9 runs ($p_{\text{adj}} = 0.38$). On the three working encoders the simulator costs 0.75–0.86 on 0 or 1 of 9 runs. The best head for AAGCN is the gated head without physics (16.57 against the direct head’s 16.71), a comparison we did not pre-register and report only as a lead.

On test the picture reverses: the SEIR head beats its no-physics twin on the failing encoders (STGAT −0.62, 8/9; A3TGCN −1.66 and DCRNN −3.50, both 9/9). That is the recurring disagreement of §5.7. It was not used to decide anything, and we do not claim it.

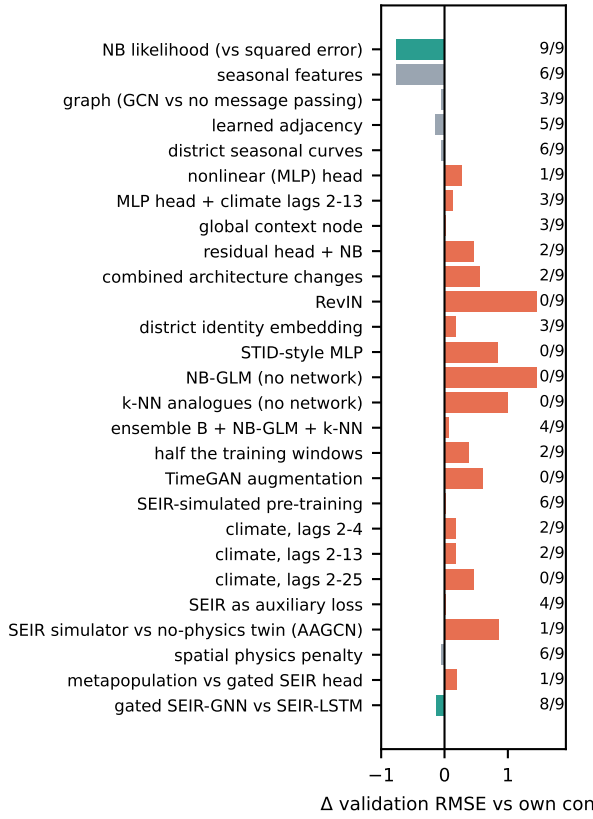


Figure 2: Every lever’s validation delta against its own control (Table 3); win counts on the right. Green: negative and $\geq 7/9$ wins. Grey: negative but inconsistent. Red: worse.

5.3 What moves the metric

Table 3 pairs every lever we tried against its own control, and Figure 2 shows the same deltas. The **negative-binomial likelihood** is the largest controlled gain: -0.76 , $9/9$, $p_{\text{adj}} = 0.014$, larger than any architecture difference. It corrects a bias, since squared error on $\log(1 + y)$ targets a median while RMSE scores a mean. **Seasonal features** give a similar mean change (-0.76) but win only $6/9$. Combined, the best model B (AAGCN, direct head, NB, seasonal features) scores 15.51 against persistence’s 17.86 : -2.35 on $9/9$ runs.

That is the headline a benchmark would report. It does not hold out of sample: on test, B scores 37.54 against persistence’s 36.02 , and the NB likelihood is *worse* on test by 1.61 . We return to this in §5.7.

5.4 SEIR-GNN against SEIR-LSTM

The comparison the research question poses is a graph encoder against the LSTM, behind the same gated SEIR head (Table 4). On three origins the graph wins on validation: ASTGCN -0.54 ($9/9$) and AAGCN -0.14 ($8/9$, $p_{\text{adj}} = 0.020$ in Table 3). Behind the direct head the same graph gains less and less consistently (-0.27 , $6/9$) and

loses on test ($+1.96$, $1/9$). So what the graph adds appears specific to estimating a transmission quantity.

The nine disjoint origins do not support a general claim. ASTGCN’s advantage holds on validation (-0.62 , $8/9$, $p = 0.012$) and vanishes on test ($+0.07$, $6/9$); AAGCN’s reverses the other way ($+0.18$ on validation, -0.30 on test, $5/9$, $p = 0.078$). Neither is significant on the metric where the other is. We therefore do not claim that the SEIR-GNN beats the SEIR-LSTM: it does so consistently on three origins on validation, and on nine origins it is not robust to the encoder or the metric.

5.5 Physics as structure: penalty and metapopulation coupling

Two further physics-informed structures were tested against B . The **spatial physics penalty** was the one significant physics gain in our earlier work on the benchmark array (-0.044 on STGAT, $57/60$ runs, reported in the Phase-2 study). On B it keeps its sign (-0.06) but wins only $6/9$; its log form is worse ($+0.08$, $2/9$). The **metapopulation SEIR head**, which couples districts inside the dynamics rather than through the initial state, is *worse* than the gated head ($+0.20$, $1/9$, $p_{\text{adj}} = 0.020$), and on nine origins it ties the SEIR-LSTM on test (-0.01 , $4/9$). The reason is in the data: on the district graph, Moran’s I of weekly log growth is -0.03 , against 0.29 for log cases. Levels are spatially clustered; transmission changes are not, so a learned coupling has nothing to encode.

5.6 The ceiling is informational

Table 3 records what did not work. No architecture change improves validation by more than 0.15 (a learned adjacency, $5/9$), and several make it worse: the nonlinear head by 0.28 , the residual head with NB by 0.47 and all changes combined by 0.55 . The literature’s remedies for weak spatio-temporal models are neutral or worse: RevIN costs 1.47 ($0/9$) and k -nearest-neighbour analogues with no network cost 1.01 ($0/9$). Data-side levers fail too. Training on a quarter of the windows costs only 0.25 , and on three-quarters 0.11 , so the model is not short of data. TimeGAN augmentation costs 0.60 ($0/9$) and truncates the tail: its largest generated week is $2,036$ cases against a real $2,631$. Climate costs 0.18 , 0.19 and 0.47 at lags $2-4$, $2-13$ and $2-25$. Gradient-boosted trees do gain from climate (-0.54 , $6/9$), but remain far behind B (17.40).

These failures share a cause. The validation residuals of the working encoders, the SEIR-LSTM and k -NN, which has no network and no training, correlate $0.91-0.97$ with B ’s: the models make the same errors. The remaining error is a lag on moves. B under-predicts rising weeks by 11.5 cases and over-predicts falling weeks by 10.1 , and the top 5% of district-weeks carry $54-67\%$ of every model’s squared error. That is what a mean-seeking objective produces on a near-random-walk target whose next move is not in the inputs.

5.7 Validation and test disagree, systematically

Across the eight families of runs, the configuration validation selects is worse than persistence on test in five, and the best configuration on test is anchored to the last observation (gated SEIR, residual, gated or k -NN) in six. Examples: the residual head with NB (34.59 against B ’s 37.54), k -NN (34.34), the SEIR-LSTM (34.97), and on nine origins the gated SEIR-GNN at 31.17 against B at 34.00 and

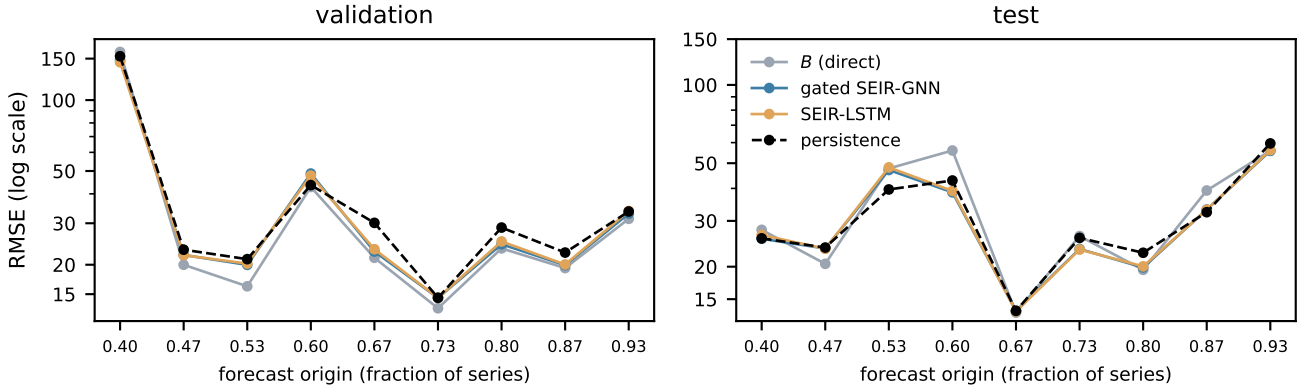


Figure 3: Nine disjoint origins, per-origin RMSE on a log scale. Origin 0.40’s validation span contains the 2017 epidemic and dominates every validation mean; read win counts, not means.

persistence at 31.76. Nothing here was selected on test, and on nine origins the gated SEIR-GNN’s test advantage over persistence is not significant ($-0.59, 6/9, p = 0.68$). A pattern across most families is nonetheless evidence about the selection protocol. A 30-week validation span immediately before each origin appears to reward fitting that span’s level, which does not carry into the next period. Acting on it needs a new pre-registered protocol and a test set nothing here has touched, which only data after March 2024 can provide.

6 DISCUSSION

Where physics helps a graph network, and where it does not. On this target the SEIR model cannot be the forecaster. A pure SEIR decoder inherits the unpredictability of its force of infection, and 14–16% of targets lie out of its reach. The gated SEIR head seemed to show that physics could still help as structure, since it made every encoder usable and repaired three published architectures. The control in §5.2 removes that reading. The repair comes from anchoring the forecast to last week’s value, which a residual head does without any mechanism, and on validation the simulator adds nothing or costs up to 0.9 RMSE. Coupling districts inside the dynamics, as metapopulation GNNs do, does not help either, because transmission has almost no local spatial structure at district-week resolution. The prior that matters here is the one the data already state, that next week looks like this week.

Two findings keep the question open rather than closed. First, on test the SEIR head beats its no-physics twin on the failing encoders (up to $-3.50, 9/9$), and every gated SEIR arm we trained sits below persistence on test, on three origins and on nine; the direct models do so only in some. Second, the graph’s advantage over the LSTM appears mainly inside the SEIR head. Neither survives our selection protocol, and we have argued that the protocol itself may be the reason (§5.7). A test of either needs data nothing here has touched.

Why the ceiling is not architectural. The target’s own last value explains $r^2 = 0.89$ of the next, the best covariate 0.03, and what remains is a move the inputs do not reveal. Every model family converges to the same function, with residual correlation 0.91–0.97

among the working models. Architecture, embeddings, climate, synthetic data and physics penalties each improve validation by at most 0.15, and most make it worse. A study that searches this space and reports its best cell will find a winner; a paired study finds noise. The NB likelihood is the exception, because it corrects what the objective targets rather than what the model sees.

What the evaluation taught us. Each of the following would have produced a publishable positive result had it gone unchecked:

- the benchmark array leaks future data through its row order and its climate channels;
- our own first SEIR-GNN pipeline reported a win that compared one origin against a three-origin baseline and did not survive a correct training loop;
- the gated SEIR head’s repair of three encoders looked like physics and was the anchor;
- a validation span before each origin selects models that underperform anchored ones on test, in five of eight families of runs;
- on nine origins, the SEIR-GNN’s advantage over the SEIR-LSTM holds on validation for one encoder and on test for the other.

Limitations. One country and one disease at district-week resolution. Three origins cannot support claims at the origin unit, and the nine-origin test spans were already examined in our earlier experiments, so they are not fresh. The STGAT spatial-penalty result we contrast with (57/60 runs) comes from the Phase-2 study on the benchmark array and is not rerun here. The graph is the benchmark’s hand-built adjacency; mobility data might couple districts differently. The SEIR head uses fixed stage durations and one initial susceptible fraction for every district.

Next steps. Clean confirmation needs data nothing here has touched: the weekly reports after March 2024. Before any such run we would pre-register a single endpoint, selection over several validation spans rather than one, and four finalists: *B*, the gated head without physics, the gated SEIR-GNN and the SEIR-LSTM.

That design would settle whether the simulator’s test-set advantage is real.

7 CONCLUSION

We asked whether replacing the LSTM of an SEIR-informed forecaster with a graph network improves district-level dengue forecasts. Under a leak-free protocol on a corrected dataset, the answer is narrower than either literature suggests. The graph adds little on its own (0.06). The graph’s advantage over the LSTM inside the SEIR head is consistent on three origins and not robust on nine. The largest controlled gain comes from the likelihood, not the architecture. Physics fails as a decoder for measurable, structural reasons, and its apparent success as a gated head belongs to the persistence anchor it is gated onto. The source-file dataset, the notebook that produces every number here, the pre-registrations and the experiment log accompany this paper.

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